Mini project Task 02 — Automatic Gate Control System (ATmega32) *[70 Marks]*

Objective:

Design and implement an automatic gate control system using an **ATmega32** microcontroller. The system must use LEDs, serial communication, interrupts, timers, PWM for servo control, and a 16×2 LCD interfacing.

System Requirements

1. **LED Indicators (STOP / GO)**
   - Use two LEDs: **Red = STOP**, **Green = GO**.
   - On system start, **only the Red LED is ON** and Green LED is OFF.
2. **Serial Command**
   - The system must receive the message **"Open Gate"** via serial communication from an external device (e.g., PC or another MCU).
3. **Serial Interrupt Handling**
   - The reception of the "Open Gate" message must be detected and handled using the **USART Receive Interrupt**.
4. **LCD Display Output (16×2)**
   - **Line 1:** After the command is received, display:
     Thank you for parking with us today!
   - **Line 2:** Display a **10-second waiting countdown**. The countdown must be implemented using **Timer0 only**.
5. **Gate Opening Mechanism (Servo Control)**
   - Use **Timer1 PWM** to control the servo motor that opens the gate.
   - When the command is received, the gate must **open** and remain open for **5 seconds**.
6. **LED Update During Open State**
   - While the gate is open, **turn ON the Green LED** and **turn OFF the Red LED**.
7. **Gate Closing**
   - After the 5-second open period, **automatically close the gate** using **Timer1 PWM**.
8. **Return to Initial State**
   - After closing the gate, the system must return to the initial state: **only the Red LED ON**, Green LED OFF, and wait for the next serial command.


Send message → " Open Gate "
EASY
COME
EASY